

<code><article></code>	• Defines an article in the document
<code><aside></code>	• Defines content aside from the page content
<code><details></code>	• Defines additional details that the user can view or hide
<code><div></code>	• Defines a div-like box or window
<code><figcaption></code>	• Defines a caption for a <figure> element
<code><figure></code>	• Defines self-contained content, like illustrations, diagrams, photos, code listings, etc.
<code><figure></code>	• Defines a holder for the document or a section
<code><h1></code>	• Defines a header for the document or a section
<code><h2></code>	• Defines navigation links in the document
<code><h3></code>	• Defines a section in the document
<code><h4></code>	• Defines a section in the document
<code><h5></code>	• Defines a visible heading for a <details> element
<code><h6></code>	• Defines a date/time

Figure 1: Semantic elements

<code><datalist></code>	• Defines pre-defined options for input controls
<code><keygen></code>	• Defines a key-pair generator field (for forms)
<code><output></code>	• Defines the result of a calculation

Figure 2: Form elements

<code><canvas></code>	• Draw graphics, on the fly, via scripting (usually JavaScript)
<code><svg></code>	• Draw scalable vector graphics

Figure 3: Graphic elements

<code><audio></code>	• Defines sound content
<code><embed></code>	• Defines containers for external applications (like plug-ins)
<code><source></code>	• Defines sources for <video> and <audio>
<code><track></code>	• Defines tracks for <video> and <audio>
<code><video></code>	• Defines video or movie content

Figure 4: Media elements

and/or browser popup. All the latest versions of Chrome, Firefox, IE, Safari and Opera can use the geo-location feature of HTML5.

Some uses of geo-location are:

- Public transportation websites
- Taxi and other transportation websites
- To calculate shipping costs on an e-commerce site
- Travel agency websites
- Real estate websites
- Movie theatre websites can find movies playing nearby
- Online gaming
- For sites to feature local headlines and weather on their front page

- Job postings can automatically include commute times
- How it works:** Geo-location works by scanning common sources of location information, which include the following:
- Global Positioning System (GPS), which is the most accurate
 - Network signals—IP address, RFID, Wi-Fi and Bluetooth MAC addresses
 - GSM/CDMA cell IDs
 - User inputs
- The API offers a very handy function to detect geo-location support in browsers:

```
if (navigator.geolocation) {
    // do stuff
}
```

The `getCurrentPosition` API is the main method for using geo-location. It retrieves the current geographic location of the user's device. The location is described as a set of geographic coordinates along with the heading and speed. The location information is returned as a position object.

The syntax is:

```
getCurrentPosition(showLocation, ErrorHandler, options);
```

- *showLocation*: This defines the callback method that retrieves location information.
- *ErrorHandler(Optional)*: This defines the callback method that is invoked when an error occurs in processing the asynchronous call.
- *options (Optional)*: This defines a set of options for retrieving the location information.

Figure 5 incorporates the set of properties returned by a position object.

We can present location information to the user in two ways—geodetic and civil:

1. The geodetic way of describing a position refers directly to the latitude and longitude.
2. The civic representation of location data is readable and easily understood by humans.

<code>coords.latitude</code>	•The latitude as a decimal number
<code>coords.longitude</code>	•The longitude as a decimal number
<code>coords.accuracy</code>	•The accuracy of position
<code>coords.altitude</code>	•The altitude in meters above the mean sea level
<code>coords.altitudeAccuracy</code>	•The altitude accuracy of position
<code>coords.heading</code>	•The heading as degrees clockwise from North
<code>coords.speed</code>	•The speed in meters per second
<code>Timestamp</code>	•The date/time of the response

Figure 5: Position object properties